Homework 5: Modern Statistics for Engineers

Completed by Brandon Auyeung for Dr. Peyman Abbaszadeh

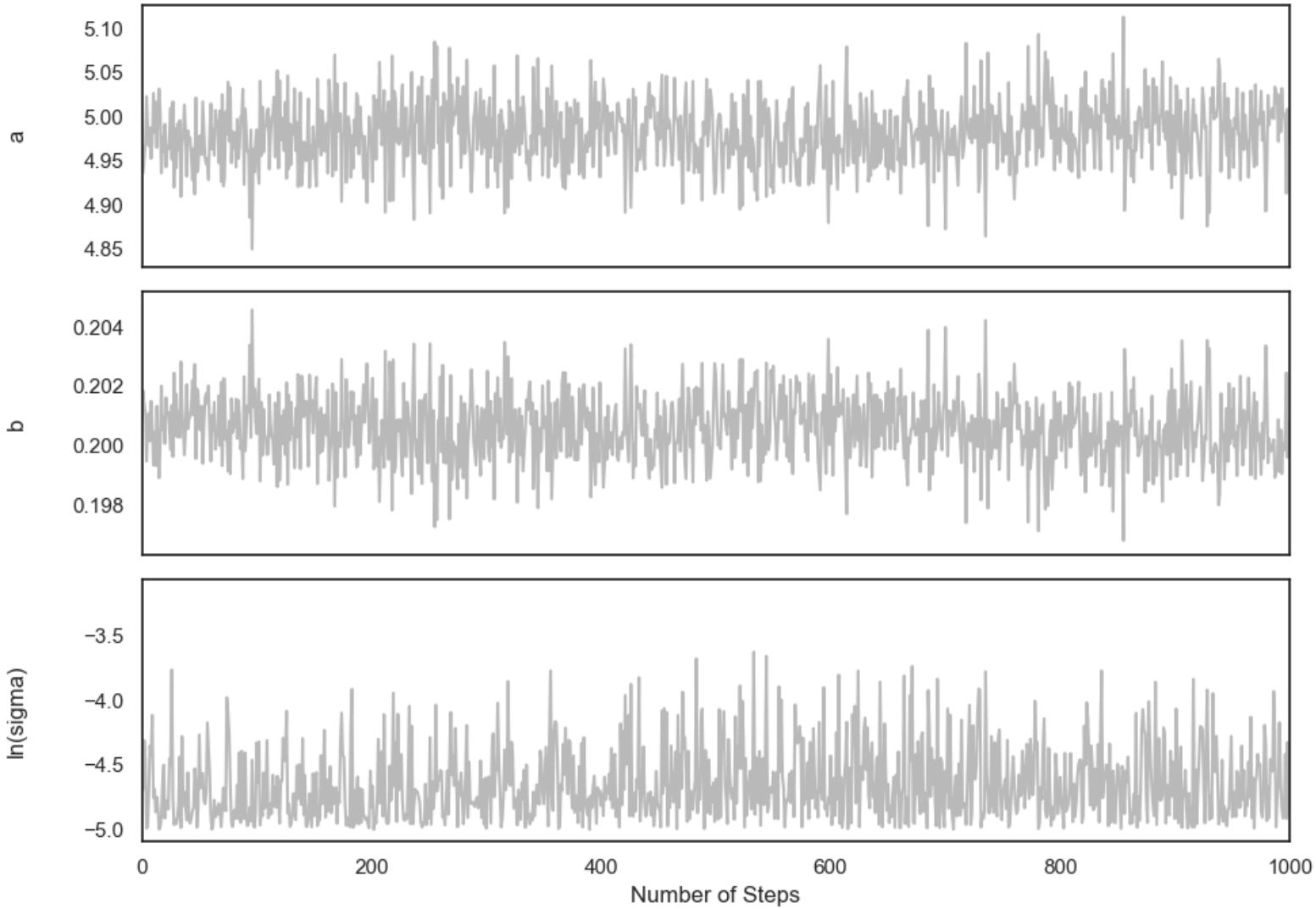
Due Date: 16th of March, 2026

1. MCMC Simulation: Use the `emcee` library to set up an MCMC simulation. Assume a linear regression model for simplicity and set up the log likelihood function. The goal is to estimate the parameters (`intercept`, `slope`, and `sigma`) that best describe the relationship between average daily temperature and heatwave durations.

In [38]:
```
np.random.seed(12)

# True parameters for generating synthetic data
true_intercept = 5
true_slope = 0.2
true_sigma = 2

avg_daily_temperature = np.random.uniform(25, 40, 1000)

# Simulate noisy heatwave durations
noise = np.random.normal(loc=0, scale=true_sigma, size=1000)
heatwave_durations = true_intercept + true_slope * avg_daily_temperature + noise

def log_prior(theta):
    intercept, slope, log_f = theta
    if not (-20.0 < intercept < 20.0 and -1.0 < slope < 1.0 and -5.0 < log_f < 5.0):
        return -np.inf
    return 0.0

def log_likelihood(theta, x, y, yerr):
    intercept, slope, log_f = theta
    f = np.exp(log_f)
    y_model = intercept + slope * x
    variance = yerr**2 + f**2
    if np.any(variance <= 0) or not np.isfinite(f):
        return -np.inf
    return -0.5 * np.sum(np.log(2 * np.pi * variance) + (y - y_model)**2 / variance)

def log_posterior(theta, x, y, yerr):
    lp = log_prior(theta)
    if not np.isfinite(lp):
        return -np.inf
    return lp + log_likelihood(theta, x, y, yerr)


# Set up the MCMC sampler
ndim = 3 # number of parameters in the model
nwalkers = 50 # number of MCMC walkers
nburn = 1000 # "burn-in" period to let chains stabilize
nsteps = 2000 # number of MCMC steps to take

initial_guess = [true_intercept, true_slope, np.log(true_sigma)]
```

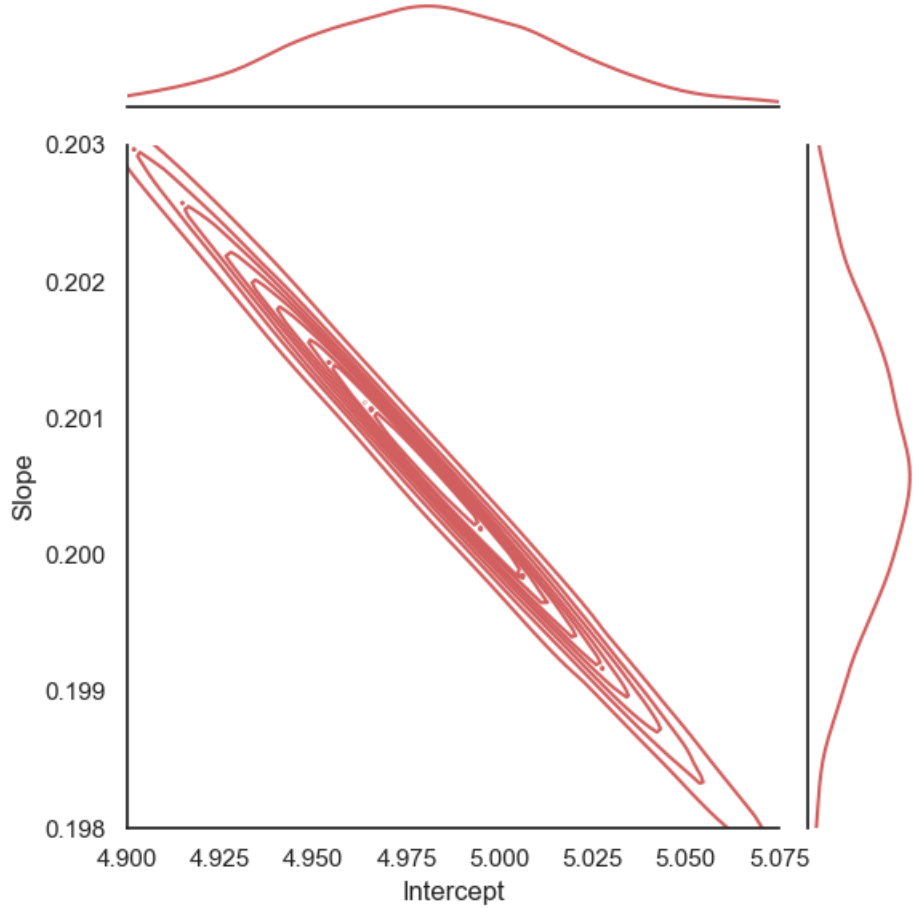
```python
initial_position = initial_guess + 1e-4 * np.random.rand(nwalkers, ndim)

sampler = emcee.EnsembleSampler(
    nwalkers, ndim, log_posterior, args=(avg_daily_temperature, heatwave_durations, noise)
)

sampler.run_mcmc(initial_position, nsteps, progress=True)

fig, axes = plt.subplots(3, figsize=(10, 7), sharex=True, tight_layout=True)
samples = sampler.get_chain(discard=nburn, thin=10, flat=True)
labels = ["a", "b", "ln(sigma)"]
for i in range(ndim):
    ax = axes[i]
    ax.plot(samples[ :, i], "k", alpha=0.3)
    ax.set_xlim(0, nsteps - nburn)
    ax.set_ylabel(labels[i])
    ax.yaxis.set_label_coords(-0.1, 0.5)

axes[-1].set_xlabel("Number of Steps");


est_intercept = np.mean(samples[ :, 0])
est_slope = np.mean(samples[:, 1])
est_sigma = np.exp(np.mean(samples[:, 2]))

print("True intercept:", true_intercept)
print("Estimated intercept:", est_intercept)
print("True slope:", true_slope)
print("Estimated slope:", est_slope)
print("True sigma:", true_sigma)
print("Estimated sigma:", est_sigma)
```

```
100%|██████████| 2000/2000 [00:03<00:00, 655.54it/s]
True intercept: 5
Estimated intercept: 4.9800687186199735
True slope: 0.2
Estimated slope: 0.20062180320132605
True sigma: 2
Estimated sigma: 0.009483753173327359
```



2. KDE Visualization: After running the MCMC simulation, extract the samples from the chain and use Kernel Density Estimation (KDE) to visualize the joint distribution of the intercept and slope parameters. Plot a 2D KDE plot to gain insights into the relationships between temperature and heatwave durations.

In [39]:
```
# # Plot the MCMC samples using KDE
sns.set_theme(style="white", palette="muted")
sns.jointplot(x=samples[:, 0], y=samples[:, 1], kind="kde", space=0, color="r")
plt.xlim(4.9, 5.075)
plt.ylim(0.198, 0.203)

plt.xlabel("Intercept")
plt.ylabel("Slope")

# Show the figure
plt.tight_layout()
```



3. Answer the following questions based on the KDE plot:

3.1. What does the joint distribution reveal about the impact of average daily temperature on heatwave durations?

It can be observed that the distribution of the slope is always positive. This means that if the average daily temperature is increased, the heatwave duration is increased.

3.2. Can you identify any patterns or clusters in the joint distribution, and what insights do they provide about the model parameters?

There is a negative correlation between the intercept and the slope. This is also a very tight distribution, which implies that the model is able to predict the heatwave duration accurately based on the average daily temperature.